Business Requirement Document (BRD)

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2 Grocery Inventory & Sales Performance Dashboard

2.1 Microsoft Excel Data Analytics Project

Project Name: Grocery Inventory & Sales Performance Dashboard

Domain: Grocery Retail & Inventory Management

Tool Used: Microsoft Excel

Project Type: Data Analytics Portfolio Project

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4 1. Project Overview

Retail grocery businesses manage hundreds of products across multiple categories while balancing inventory availability, sales performance, and replenishment planning. Poor inventory management can result in stock shortages, excess inventory, increased holding costs, and revenue loss.

The objective of this project is to develop an interactive Microsoft Excel dashboard that provides a centralized view of inventory and sales performance. The dashboard enables business users to monitor key performance indicators (KPIs), analyze category-wise performance, track monthly revenue trends, identify products requiring replenishment, and support data-driven decision-making.

The dashboard combines data cleaning, calculated columns, Pivot Tables, Pivot Charts, KPI cards, slicers, and timeline filtering to transform raw operational data into meaningful business insights.

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5 2. Business Problem

The grocery business maintains large volumes of inventory and sales data. Although the data is available, it does not directly provide meaningful insights for management.

Business users face several operational challenges:

- Difficulty identifying top-performing product categories.
- No centralized reporting solution for inventory and sales.
- Limited visibility into current inventory value.
- Difficulty identifying products requiring replenishment.
- Limited understanding of inventory turnover efficiency.
- Inability to analyze monthly revenue trends.
- Time-consuming manual report preparation.

Without an interactive dashboard, management spends considerable time preparing reports and making decisions based on raw Excel data.

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6 3. Business Objectives

The primary objective of this project is to transform raw inventory and sales data into an interactive dashboard that enables business users to monitor operational performance and make informed business decisions.

The dashboard aims to:

- Monitor overall business performance.
- Analyze total revenue generated.
- Evaluate current inventory value.
- Track monthly revenue trends.
- Identify products requiring replenishment.
- Measure inventory turnover efficiency.
- Compare revenue across product categories.
- Identify top revenue-generating products.

- Provide interactive filtering for detailed analysis.
- Support data-driven inventory planning.

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7 4. Stakeholders

The dashboard is designed for multiple business users.

Stakeholder	Purpose
Business Owner	Monitor overall business performance
Store Manager	Monitor inventory and sales
Inventory Manager	Track stock levels and inventory value
Sales Manager	Analyze sales performance and revenue
Procurement Team	Plan product replenishment
Data Analyst	Generate reports and business insights

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8 5. Dataset Description

The dataset contains inventory and sales information for grocery products across multiple categories.

8.0.1 Dataset Summary

- Domain: Grocery Retail
- Total Products: **990**
- Categories: **7**
- Analysis Period: **2024–2025**

8.0.2 Major Fields

- Product_ID
- Product_Name
- Category
- Supplier_ID
- Supplier_Name
- Stock_Quantity
- Reorder_Level
- Reorder_Quantity
- Unit_Price
- Sales_Volume
- Inventory_Turnover_Rate
- Date_Received
- Last_Order_Date
- Expiration_Date

8.0.3 Calculated Fields

- Revenue
- Inventory_Value
- Reorder_Status
- Month_Year

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9 6. Business Requirements

The dashboard should answer the following business questions.

9.1 Sales Analysis

- What is the total revenue generated?
- Which product category contributes the highest revenue?
- Which products generate the highest revenue?
- How does revenue change month by month?

9.2 Inventory Analysis

- What is the current inventory value?
- Which category has the highest inventory value?
- Which products require replenishment?
- What percentage of products require reorder?

9.3 Product Performance

- Which are the Top 10 revenue-generating products?
- Which categories have the highest inventory turnover?

9.4 Operational Performance

- How efficiently is inventory moving?
- Which products require immediate replenishment?
- How does business performance change across different years?

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10 7. Functional Requirements

The dashboard shall provide:

- Interactive KPI Cards
- Dynamic Pivot Tables
- Dynamic Pivot Charts
- Monthly Revenue Trend
- Category Slicer
- Reorder Status Slicer

- Year Timeline Filter
- Automatic dashboard updates when filters are applied
- Interactive filtering across all dashboard visualizations

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11 8. Key Performance Indicators (KPIs)

The dashboard displays the following KPIs.

KPI	Description
Total Products	Total number of products
Total Revenue	Total revenue generated
Total Units Sold	Total quantity sold
Inventory Value	Current inventory value
Reorder Products	Number of products requiring replenishment
Reorder Percentage	Percentage of products requiring reorder
Average Inventory Turnover Rate	Average inventory turnover efficiency

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12 9. Dashboard Visualizations

The dashboard contains the following visualizations.

12.1 KPI Cards

- Total Products
- Total Revenue
- Total Units Sold
- Inventory Value
- Reorder Products
- Reorder Percentage
- Average Inventory Turnover Rate

12.2 Charts

12.2.1 Monthly Revenue Trend

Chart Type: Line Chart

Purpose

Displays monthly revenue performance for the selected year, helping users identify revenue trends and seasonal patterns.

12.2.2 Revenue by Category

Chart Type: Clustered Column Chart

Purpose

Compares revenue generated by each grocery product category.

12.2.3 Inventory Value by Category

Chart Type: Horizontal Bar Chart

Purpose

Displays the monetary value of inventory held across different categories.

12.2.4 Top 10 Products by Revenue

Chart Type: Horizontal Bar Chart

Purpose

Ranks the ten highest revenue-generating products.

12.2.5 Inventory Turnover Rate by Category

Chart Type: Clustered Column Chart

Purpose

Compares inventory turnover efficiency across product categories.

12.2.6 Reorder Status Distribution

Chart Type: Doughnut Chart

Purpose

Shows the proportion of products requiring replenishment versus products with sufficient stock.

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13 10. Dashboard Filters

The dashboard includes interactive filters.

Filter	Purpose
Category	Analyze data by product category

Filter	Purpose
Year Timeline	Analyze monthly revenue by year
Reorder Status	Filter products based on stock availability

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14 11. Business Rules

The following business rules were implemented.

14.0.1 Revenue

```
Revenue = Sales Volume × Unit Price
```

14.0.2 Inventory Value

```
Inventory Value = Stock Quantity × Unit Price
```

14.0.3 Reorder Status

```
If Stock Quantity < Reorder Level

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Reorder Required

Otherwise

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Sufficient Stock
```

14.0.4 Reorder Percentage

```
Reorder Percentage =
(Reorder Products ÷ Total Products) × 100
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15 12. Assumptions

The following assumptions were made during development.

- The dataset represents a grocery retail business.
- Currency values are represented in USD.
- Revenue is calculated using Sales Volume and Unit Price.
- Inventory Turnover Rate is assumed to be pre-calculated.
- Date_Received is used for monthly trend analysis.

- All date fields were standardized before analysis.

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16 13. Project Scope

16.1 Included

- Data Cleaning
- Data Transformation
- KPI Development
- Calculated Columns
- Pivot Tables
- Pivot Charts
- Dashboard Design
- Interactive Slicers
- Timeline Filter
- Monthly Revenue Trend
- Business Reporting

16.2 Excluded

- Profit Analysis
- Machine Learning
- Sales Forecasting
- SQL Integration
- Power BI Dashboard
- Real-time Data Refresh

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17 14. Limitations

The following limitations were identified during the project.

- Profit and cost information were unavailable.
- Supplier analysis was not included.
- Historical data covers only 2024–2025.
- Real-time inventory tracking is outside the project scope.
- Expiry risk analysis was not included due to dataset limitations.

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18 15. Expected Business Benefits

The completed dashboard enables business users to:

- Monitor business performance from a single dashboard.
- Analyze monthly revenue trends.
- Improve inventory planning.

- Reduce stock shortages.
- Identify high-performing products.
- Compare category performance.
- Monitor inventory turnover.
- Support faster, data-driven decision-making.
- Reduce manual reporting effort.

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19 16. Conclusion

The **Grocery Inventory & Sales Performance Dashboard** successfully transforms raw inventory and sales data into an interactive business intelligence solution using Microsoft Excel.

The dashboard integrates data cleaning, calculated columns, KPI development, Pivot Tables, Pivot Charts, slicers, and timeline filtering to provide a comprehensive view of business performance.

By combining inventory analysis, sales performance, monthly revenue trends, and operational KPIs, the dashboard enables managers to monitor business health, identify opportunities for improvement, and make informed decisions.

This project demonstrates practical skills in **data cleaning, business analysis, dashboard development, KPI design, and data visualization**, making it a strong portfolio project for aspiring Data Analysts.